

Capstone controlled experiments

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone hypothesis and baseline

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Efficiency and resource measurement

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Transfer learning

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace both runs for Capstone controlled experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Capstone controlled experiments. Proposed explanation: “More runs mean more validation examples”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Run a new comparison independently

For training (1,3),(2,6) and validation (3,9),(4,12), start $a = 1$. Compare rates 0.1 and 0.2 for two updates. Gradient is $5*(a-3)$. Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Name the controls

List six settings held fixed and the single intended changed factor.

My reasoning and evidence

6. W2 Trace both runs

Compute both updates for rates 0.1 and 0.2 using gradient $5*(a-2)$.

My reasoning and evidence

7. W3 Select without extra evidence

Recompute validation MSE for the final weights and apply the written selection rule. How many distinct validation rows were used?

My reasoning and evidence

8. W4 Run a new comparison independently

For training (1,3),(2,6) and validation (3,9),(4,12), start $a = 1$. Compare rates 0.1 and 0.2 for two updates. Gradient is $5*(a-3)$. Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

My reasoning and evidence

9. W5 Audit a mixed comparison

Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different training rows. Can the difference be attributed to rate alone? Propose a repair.

My reasoning and evidence

10. Project checkpoint and reflection

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final evidence pack: For training (1,3),(2,6) and validation (3,9),(4,12), start $a = 1$. Compare rates 0.1 and 0.2 for two updates. Gradient is $5 \cdot (a-3)$. Compute final weights and validation MSE, then select lower MSE; ties go to 0.1. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Capstone controlled experiments

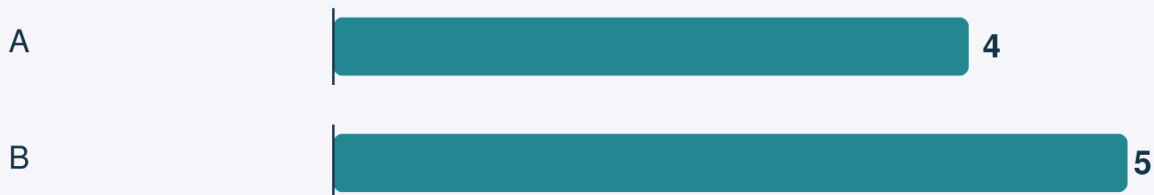
One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1]. Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

AI ACADEMY / AI-5/34

Controlled toy-pattern comparison

correct labels out of 6 shared patterns



Bar length shows the supplied value; the marked baseline is zero.

Capstone controlled experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the intervention

B adds the centre-feature connection; all other declared settings stay fixed.



2. Count outcomes

A is 4/6 correct; B is 5/6.



3. Inspect paired cases

Only case three changes correctness; case five fails for both.

CHANGE ONE THING

Change B's training set as well as the feature connection.

Explain the changed result using the matching lesson.

Compute the two correct counts and identify the changed case. Why would changing B's training data weaken the conclusion about its added connection?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.